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Weather sensitivity of bike-share usage a hierarchical Bayesian plan for 25 cities

STA5001 Bayesian & Computational Statistics

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Weather associations in bike-share usage

The reference study. Bean, Pojani and Corcoran (2021) compare **forty cities** across climate zones, fitting a separate generalized additive model to each.

Two robust patterns: hires rise with UTCI up to about **27–28 °C** and then fall; rainfall is associated with lower use.

UTCI combines temperature, humidity, wind and radiation to describe thermal stress.

Why estimate cities jointly?

- About one year of daily observations per city
- Shared estimation of city weather coefficients and their spread

Question. Are city thermal and rainfall backgrounds associated with weather sensitivity?

Data: one public release, 29 reproducible cities

Source	
Release	Mendeley Data, CC BY 4.0
DOI	`10.17632/2nxpvtz935.1`
Content	hourly hires, UTCI, rainfall
Scope	29 dock-based systems

Why 29 and not 40. The paper's own code designates the dock-based systems; the other cities have metadata but no hourly stock data.

Each city contributes exactly **8,760** hourly rows.

Archive and paper are both stored with SHA-256 digests, re-checked on every run.

From hourly records to 9,120 analysis days

Processing

- Preserve row order, map onto consecutive UTC hours
- Convert to the city's local time zone
- Aggregate by local date

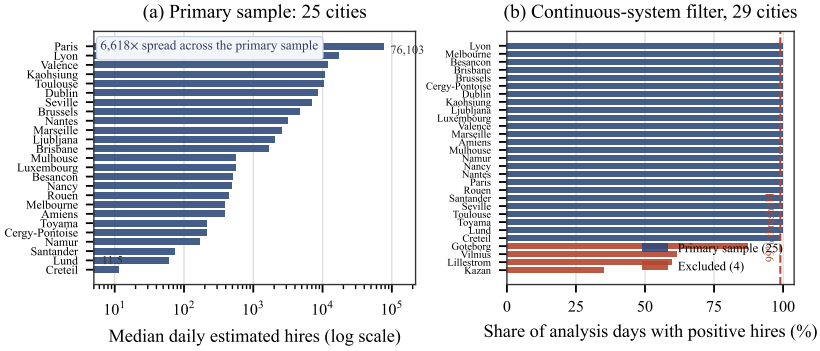
Quality rules

- 23–25 hourly rows per local day
- at least 20 valid UTCI hours
- complete precipitation

254,040 hourly rows → 10,613 candidate city-days → **10,578** kept → **9,120** in the primary sample of 25 cities

Three repeated time keys exist; Dublin and Melbourne coincide with daylight-saving transitions, Brisbane does not observe daylight saving; the paper's row-order rule is kept.

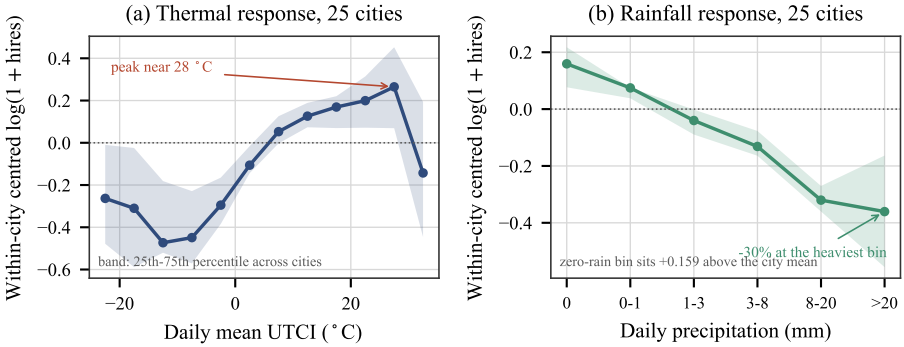
Cities differ in scale by four orders of magnitude



Median daily hires run from **11.5** in Créteil to **76,103** in Paris.

The four cities below the 99% line are excluded from the primary sample.

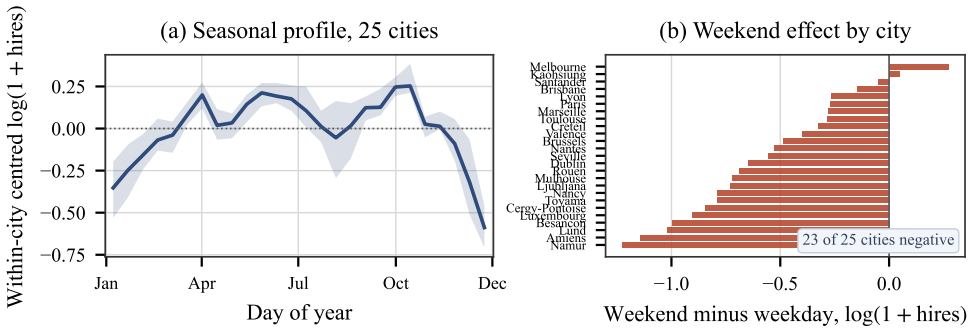
Thermal response peaks near 27.5 °C



Peaks at **+0.26** near **27.5 °C**, then falls to -0.14 in the hottest bin.

Rainfall declines monotonically: the heaviest bin sits about **41%** below the dry bin.

Season and week disagree across cities



The seasonal range is **0.84** log units; weather estimates require seasonal adjustment.

23 of 25 cities have lower weekend use; Melbourne is the only clearly positive city.

Plan: jointly estimate city weather associations

Response. $y_{it} = \log(1 + U_{it})$, Student-$t$ observation distribution with estimated degrees of freedom.

Location. city intercept, thermal anomaly and its square, standardised rainfall, weekend, annual sine and cosine.

Hierarchy. city coefficients share a normal population distribution; weather coefficients use a non-centred form.

What this buys

- Between-city spread is an estimated quantity
- Population means vary with city climate background
- City estimates share information

NUTS via PyMC and nutpie; multiple chains and convergence checks.

Take-away

1. A reproducible panel of **25 cities and 9,120 city-days** is frozen and documented.
2. The thermal response is **non-monotone**, peaking near **27.5 °C**, and rainfall is associated with monotonically lower use.
3. Cities differ in scale, in weekend behaviour and in their weather response, motivating **city coefficients with a shared distribution**.

Questions

Backup: filtering and sample construction

Stage	Result
Hourly stock rows for 29 cities	254,040
Candidate local days	10,613
Days failing a quality rule	35
Retained city-days	10,578
Cities recording hires on $\geq 99\%$ of days	25
Primary sample	9,120 city-days

Exclusions overlap: 28 days outside the nominal study period, 34 outside the 23–25 hour count, 33 short of 20 UTCI hours, none on precipitation. The four excluded cities are Göteborg, Kazan, Lillestrøm and Vilnius.

Backup: model and priors

Priors on the $\log(1+U)$ scale

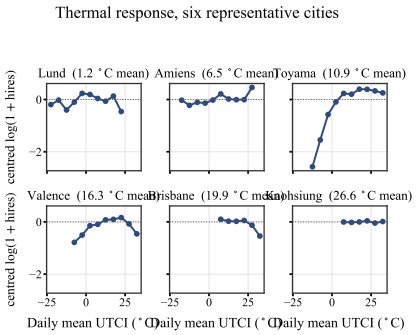
Population intercept	$\mathcal{N}(7, 2)$
City intercept scale	HN(2)
Thermal, per 10 °C	$\mathcal{N}(0, 0.5)$
Quadratic	$\mathcal{N}(0, 0.3)$
Rainfall	$\mathcal{N}(0, 0.3)$
Weekend	$\mathcal{N}(0, 0.4)$
Seasonal	$\mathcal{N}(0, 0.8)$

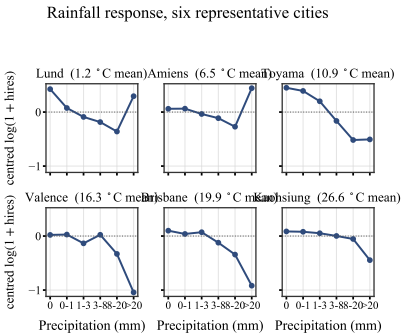
Degrees of freedom $\nu = 2 +$ Exponential(0.1); moderator coefficients use $\mathcal{N}(0, 0.2)$ to $\mathcal{N}(0, 0.3)$, and city scales use HalfNormal priors matched to the population scale.

Prior predictive checks will assess the implied ranges of log usage and weather contrasts.

Convergence thresholds: $\widehat{R} \leq 1.01$, bulk and tail ESS at 400, no divergences.

Backup: per-city curves for six representative cities





Lund (1.2 °C), Amiens (6.5), Toyama (10.9), Valence (16.3), Brisbane (19.9) and Kaohsiung (26.6) are equally spaced over the range of city means; the grids for all 25 cities are in the supplementary figures.

Backup: source and reproducibility

- Data release: Mendeley Data, DOI `10.17632/2nxpvtz935.1`, CC BY 4.0, Version 1.
- Reference paper: Bean, Pojani and Corcoran (2021), *Journal of Transport Geography* 95:103155.
- Preparation: `python code/prepare_bikeshare_weather.py`
- Daily panel: `data/clean/bikeshare_weather_daily.csv`, 10,578 rows $\times$ 28 columns.
- Data audit: `results/bikeshare_weather_data_audit.json`